

When MWPM Works - and When It Fails

A simple guide to surface-code decoding, error chains, and failure modes

Main idea: MWPM does not need to reconstruct the exact physical error. It only needs to choose a correction that puts the logical state in the correct equivalence class.

Think of MWPM as an error detective. Stim tells it which detector events appeared. MWPM compares possible error chains and chooses the lowest-cost explanation according to the decoder weights.

1. What MWPM actually sees

MWPM sees detector events, not the hidden physical fault itself. A detector event means a stabilizer result changed between rounds.

physical fault -> detector events -> possible error chains -> MWPM chooses lowest total weight -> correction / Pauli-frame update

Important: the chosen path can differ from the actual physical path and still be a successful correction, as long as their combination does not create a logical operator.

2. Situations where MWPM is likely to succeed

Situation	Why MWPM can succeed	Main reason
No physical error	No detector events appear, so no correction is needed.	No fault occurred.
One isolated local error	It usually creates a small, recognizable detector pattern.	Sparse local faults are easy to explain.
One measurement error	It often produces a short time-like detector pattern.	The graph includes measurement-fault possibilities.
One reset error	It creates a pattern tied to a known circuit location.	The fault model contains reset locations.
Short error chain	Nearby detector endpoints usually favor the correct equivalence class.	The chain is small relative to the code distance.
Several well-separated errors	Each local pattern can often be matched independently.	Patterns do not overlap strongly.
Error near a valid	The event can be matched to a boundary	Boundary geometry is

boundary	node.	part of the decoding graph.
Low physical error rate	The true low-weight explanation is usually more probable than long alternatives.	The code is operating below its useful regime.
Well-calibrated noise model	Decoder weights rank candidate paths realistically.	Assumed probabilities resemble the actual hardware/simulation.
Degenerate but logically equivalent path	A different physical route can still preserve the same logical information.	Exact path recovery is not required.

3. When MWPM is more likely to fail

MWPM fails when the lowest-weight explanation in its model is not logically equivalent to the true error. This can happen because the detector pattern is ambiguous, the noise is too strong, or the decoder weights do not describe the real noise well.

Failure situation	What is happening	Why MWPM can be wrong
Many nearby errors / overlapping patterns	Several faults produce detector events in the same region. Different error chains can explain almost the same pattern.	MWPM may pair the wrong detector events or choose the wrong equivalence class.
Error chain approaches or spans code distance	The fault chain becomes long enough to connect relevant boundaries or mimic a logical string.	A correction that looks locally reasonable can complete a logical operator.
Errors connecting opposite boundaries	Boundary matching becomes ambiguous when a chain effectively crosses the patch.	The decoder may choose the wrong boundary connection.
Several candidate paths have similar total weight	The syndrome does not strongly favor one explanation.	A tiny weight difference can push the decoder toward the wrong path.
Measurement errors are frequent	False syndrome changes appear in time and can imitate real data errors.	The space-time graph becomes crowded and ambiguous.
Temporally correlated noise but IID weights	Errors at nearby times are more related than the graph assumes.	MWPM ranks time-connected explanations incorrectly.
Leakage	A qubit can leave the computational subspace and disturb several later checks.	The resulting long streaks are not well described by simple pairwise Pauli faults.
Crosstalk / multi-qubit correlated faults	One physical disturbance can affect several qubits at once.	A single fault may create detector patterns that do not fit independent-edge assumptions.
Calibration drift	The true gate/readout/reset error rates change over time.	Static decoder weights become stale.
Physical error rate too high	Too many faults occur before the code can separate them cleanly.	Logical-error chains become common and syndrome information becomes ambiguous.

Hyperedge-like correlated fault not represented well	One fault can flip 3 or more detectors.	A simple graph decomposition can lose information about the joint event.
Wrong model for spatial or temporal structure	Decoder assumes one geometry/probability structure while hardware behaves differently.	The “cheapest” path in the graph is not the most probable path in reality.

4. The most important failure mechanism

Decoder failure = true error chain + decoder correction forms a nontrivial logical operator.

A decoder can make a physically wrong guess but still succeed. It fails only when the correction changes the logical information.

Actual path A + correction path B -> harmless loop = SUCCESS
Actual path A + correction path C -> logical string across code = FAILURE

5. Why temporal correlation matters

Under IID noise, candidate faults are weighted using independent error probabilities. If the real hardware has memory, that ranking can be wrong.

detector history -> measure temporal correlation -> estimate which time-separated faults are more plausible -> update probabilities/weights -> rerun MWPM -> compare logical error rate

Example idea: if detector events two rounds apart are much more likely to belong to the same underlying noisy episode than the IID model predicts, a time-connected path should become cheaper in the decoder graph. The correlation value itself is not directly an MWPM weight; it first has to be converted into a conditional fault probability or another principled likelihood model.

6. Simple checklist: when should I trust MWPM more?

- Physical error rate is low enough that local patterns remain sparse.
- Noise is mostly local and close to the model used to build the graph.
- Measurement and reset errors are represented in the detector error model.
- Correlated faults are weak or explicitly modeled.
- Graph weights are calibrated to realistic probabilities.
- Code distance is large relative to typical error chains.

Bottom line: MWPM is strongest when its graph is a good probabilistic picture of the real space-time error process.